

The empirical recurrence for OEIS A304547, recovered from its tail

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Abstract

OEIS A304547 records “Empirical recurrence of order 70” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 73$. Its last coefficient is $c_{70} = 1152 \neq 0$, so no further tail satisfies anything shorter; any order-70 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A304547 is “Number of $n \times 4$ 0..1 arrays with every element unequal to 0, 2, 3, 4 or 6 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

1, 24, 60, 418, 2186, 12281, 72713, 423041, ...

The entry states:

Empirical recurrence of order 70 (see link above).

As of the “Last modified” line on the live entry (Dec 10 2025 08:42:55, revision 9) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 70: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 70.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 3$ on, it returns order 70 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{70} c_i a(n-i),$$

c_1	8	c_{19}	−2289974	c_{37}	332092656	c_{55}	97889960
c_2	−7	c_{20}	26663	c_{38}	289836002	c_{56}	136379350
c_3	0	c_{21}	−320984	c_{39}	−450316588	c_{57}	−10714283
c_4	−274	c_{22}	4901063	c_{40}	113446611	c_{58}	−59513824
c_5	283	c_{23}	3845458	c_{41}	328206053	c_{59}	−3677681
c_6	967	c_{24}	7149710	c_{42}	−1211455952	c_{60}	8704362
c_7	2601	c_{25}	15330261	c_{43}	400466636	c_{61}	−35758
c_8	−3427	c_{26}	−41008590	c_{44}	1111615440	c_{62}	559358
c_9	−13714	c_{27}	36365262	c_{45}	−237025300	c_{63}	59392
c_{10}	−14178	c_{28}	−123906603	c_{46}	−810382768	c_{64}	−43500
c_{11}	19594	c_{29}	−11174793	c_{47}	176809295	c_{65}	81272
c_{12}	88273	c_{30}	−53221395	c_{48}	118893040	c_{66}	−54228
c_{13}	29353	c_{31}	98320859	c_{49}	−199417001	c_{67}	−21916
c_{14}	−55437	c_{32}	107448016	c_{50}	374755065	c_{68}	−3128
c_{15}	−296082	c_{33}	73144099	c_{51}	57739403	c_{69}	1248
c_{16}	308133	c_{34}	445281015	c_{52}	−260206991	c_{70}	1152
c_{17}	87017	c_{35}	−429631080	c_{53}	−135332725		
c_{18}	200029	c_{36}	−108848212	c_{54}	13486059		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 73$, and it is the recurrence OEIS A304547 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{70} - c_1 x^{70-1} - \dots - c_{70}$. Its constant term is $-c_{70} = -1152$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 70 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 70, so $\deg q = 70$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 70, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 24 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 70, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1152.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-70 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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